Integrative genetic, functional genomic, and single-cell transcriptomic analyses identified CXCR6 as a marker gene in COVID-19 severity

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Background

- The coronavirus disease 2019 (COVID-19) pandemic has already infected over 1.6 million people and caused numerous morbidities and over 3.4 million death worldwide as of May 2021. A huge disparity in the severity of symptoms in different patients has been observed. Several genome-wide association studies (GWAS) have been carried out, and one genomic risk locus, *3p21.31*, has been replicated to be associated with the critical illness (The Severe Covid-19 GWAS Group 2020) [1]. However, there is a cluster of six genes (*SLC6A20*, *LZTFL1*, *CCR9*, *FYCO1*, *CXCR6*, and *XCR1*) nearby the lead SNP rs35081325 within a complex linkage disequilibrium (LD) structure, which makes the "causal" gene and functional implication of this locus remain elusive[1]. In this study, we aimed to connect the genetic factors with the context-specific molecular phenotype in COVID-19 patients.

Methods

- We used two independent COVID-19 GWAS dataset Severe COVID-19 GWAS group ($GWAS_{SCGG}$) [1] and COVID-19 Host Genetics Initiative ($GWAS_{HGI}$) [2] to perform transcriptome-wide association study (TWAS) and colocalization analyses with a broad collection of eQTL datasets at the tissue and cellular levels.
- We further integrated the BALF single-cell transcriptome dataset[3] to explore the cellular transcriptome alterations in severe and moderate COVID-19 patients.

Results

Figure 1. Workflow of a data-driven study: from genetic factor to molecular phenotype.

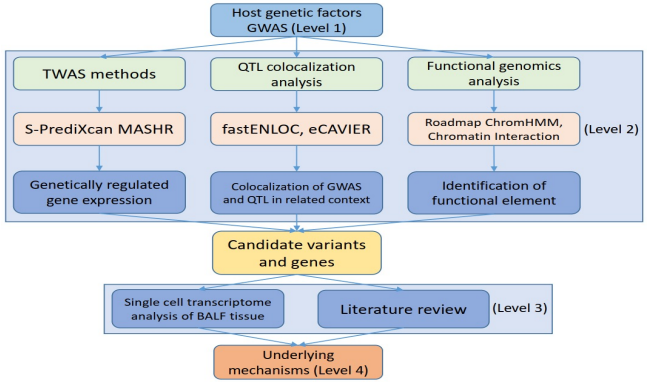


Figure 2: *CXCR6* has a protective effect in lung and whole blood

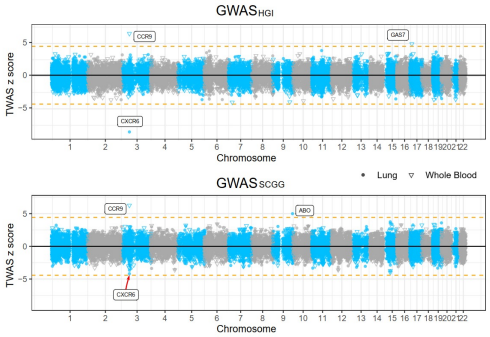


Figure 4. *CXCR6* has lower expression in severe CD8⁺ T_{RM} cells

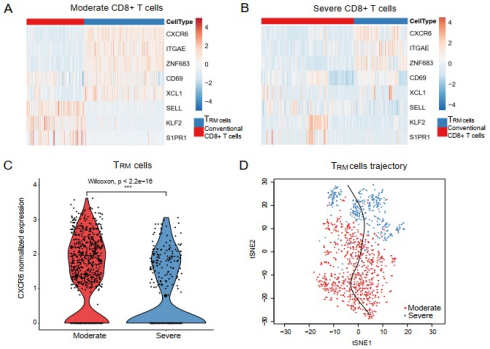
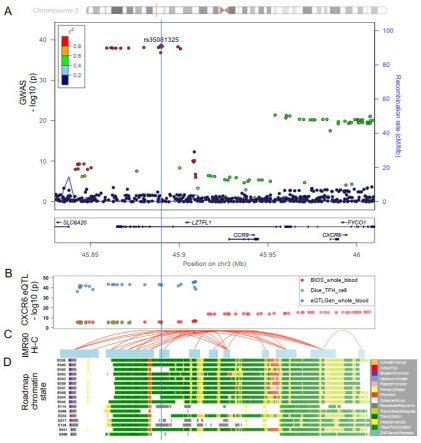


Figure 4. Single-cell transcriptome analysis of the severe and moderate COVID-19 patients. (A) Relative expression of the lung resident memory CD8⁺ T (T_{RM}) signature genes in T_{RM} cells and conventional CD8⁺ T cells in moderate patients *CXCR6* is one of T_{RM} marker genes. (B) Relative expression of the T_{RM} featured genes in T_{RM} cells and conventional CD8⁺ T cells in severe patients. (C) *CXCR6* expression in the T_{RM} cells of moderate and severe patients. (D) Pseudotime inference for the moderate and severe T_{RM} cells by Slingshot.

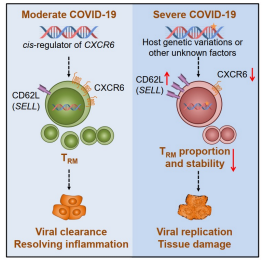
Figure 2. Manhattan plots illustrating the z scores of transcriptome-wide association study (TWAS) genes. TWAS z scores for two genome-wide association study (GWAS) datasets of susceptibility to severe COVID-19 using lung and whole blood tissue models. The upper panel shows the results from $GWAS_{HGI}$ and the lower panel from $GWAS_{SCGG}$ (see Methods). Dashed horizontal lines denote the Bonferroni-corrected significance threshold ($|z| = 4.56$, $p < 5 \times 10^{-6}$). Significant genes were highlighted with their gene symbol.

Figure 3. Functional genomic annotation in the *3p21.31* locus with signals from $GWAS_{HGI}$.
(A) LocusZoom view of the association signals of SNPs in the *3p21.31* locus of $GWAS_{HGI}$. (B) The significant eQTLs associated with *CXCR6* expression in this region. (C) The significant Hi-C interactions in normal lung fibroblast cell line (IMR-90). (D) The region annotated with the chromatin-state segmentation track (ChromHMM) from the Roadmap Epigenomics data for T-cell and lung tissue.

Figure 3. Functional genomic annotation in the *3p21.31* locus



Conclusions



1. Systematic TWAS and QTL colocalization analyses pinpointed the association of *CXCR6* with COVID-19 severity in the *3p21.31* locus.
2. Several genome-wide significant SNPs were within the LD block of *CXCR6* eQTLs in immune-related cells. In a scRNA-seq COVID-19 BALF dataset, we characterized that *CXCR6* (a T_{RM} cells marker gene) had a lower expression in the severe patients than the moderate patients.
3. The T_{RM} cells in the severe patients had a 2.24-fold proportion decrease in CD8⁺ T cells and much higher pro-inflammatory activity than those in the moderate patients.
4. We hypothesized a mechanism on how the lower expression of *CXCR6* regulated by the endogenous factors or other unknown factors could progress to severe COVID-19 outcomes.

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